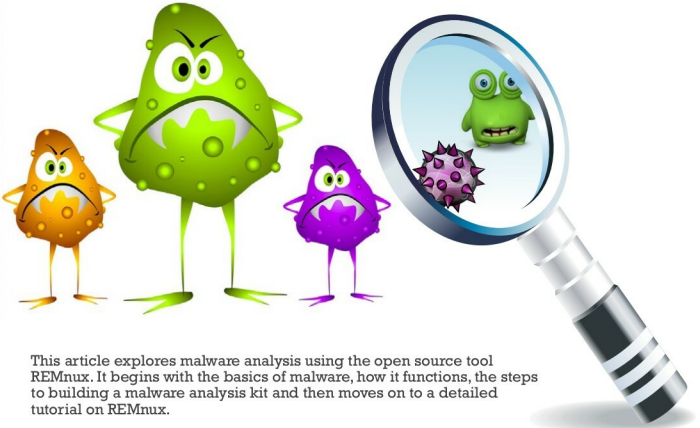


Malware Analysis Using REMnux



This article explores malware analysis using the open source tool REMnux. It begins with the basics of malware, how it functions, the steps to building a malware analysis kit and then moves on to a detailed tutorial on REMnux.

‘**M**alware is an intrusive software which includes computer viruses, worms, Trojan horses, ransomware, spyware, adware, scareware...’ – *Wikipedia*

In laymen's terms, malware is defined as any code that performs hazardous activity to the computer. It might be an executable with unknown functionality.

How malware functions

It is target-specific: Today's zero day exploits are very specific to a target and are hard to identify using traditional security solutions.

Data exfiltration: Data leakage/data extrusion is unauthorised transfer, copying or retrieval of data from a computer or server.

The Remote Administration Tool (RAT)

A RAT is malicious software that hides itself and gives full control to the attacker, remotely. Generally, a RAT gives access to the system, whereby the attacker can perform such remote activities as triggering the camera, installing rootkits and key loggers, and even grab screenshots or turn a system on/off.

General behaviour

Generally, malware hides itself and runs in the background monitoring all the processes, while communicating with the command and control server outside the network. The behaviour of malware is target-specific and influenced by the following four major characteristics.

Initial infection vector: Initial infection vector is defined as how the malware reaches the system in the first place —delivery can be through a USB stick, browser based downloads, any shared pools, etc.

Rubric: The actions the malware carries out after infection and the footprint it leaves in the log are studied via dynamic malware analysis, but zero day exploits can be unique in nature and remain undetected.

Proliferation mechanism: Generally, propagation of such malicious software depends upon what type it is—if it is a worm, it exploits a known vulnerability. If it is a rootkit, it infects files at the root of the drives and adds *autorun*. *ini*. If it is a Trojan, it hides itself or binds itself to any *dll* in system 32 and adds to the startup process. Hence, learning the propagation mechanism of a given type of malware is very important during analysis.